Case 03 · Architecture chapter: Roll Transparency Log

Following chapter 3 (architecture) of the Magshimim project-book template: plots, components, messages, a scenario in motion, failure modes, decisions, build order and storage. Characters are fictional.

Introduction

The spec described parts: capture, codes, identifiers, log, checkpoints, witnesses. But an idea runs on no computer. The builder's question: which programs exist, and where does each one sit: at the Commission, with the public, with the witnesses, or at the tribunal? And the most important question: what happens when the HSM fails on draft-publication morning?

Plots

Plot	What runs there
Commission	Roll database, change capture, reason-code service, HSM, log writer, tile store, reconciler. Only here can anyone write, and nobody here can delete.
Public	Check website, SMS gateway, kiosks, CDN. Everyone reads, nobody writes.
Witnesses	Newspapers, NGOs, parties, universities. They check every minute that the log only grows.
Tribunals	Appeal intake keyed by event id.

Components

Component	Plot	Role	Never
Roll database (PostgreSQL)	Commission	The existing system; officers add, remove and move as today	Knows it is being read; is not modified
Change capture (Debezium, Kafka)	Commission	Reads each new row and publishes a ChangeEvent	Writes to the database
Reason-code service	Commission	Checks the code is one of 11 and the evidence is attached	Decides whether a deletion is right
HSM	Commission	Number → pseudonym; signs heads	Lets a key out
Log writer (Tessera)	Commission	Appends leaves, checkpoints every minute	Changes an existing leaf
Tile store (S3)	Commission	Static tiles of 256 hashes	Deletes a tile
Check service (FastAPI)	Public	Number + date of birth → pseudonym → lines and proofs	Stores the request
Reconciler	Commission	Rebuilds the roll from the log nightly and compares	Fixes on its own
CDN, PWA, SMS	Public	Serve tiles; self-check; alerts	Hold an elector number
Witnesses (tlog-witness)	Witnesses	Check consistency and cosign	Sign a head that doesn't extend the previous one
Aggregates (Grafana)	Witnesses	District/week counts	Show a booth
Appeal intake	Tribunals	Appeal with event id	Accept an appeal without a line

Scenario: one deletion

An officer removes Priya from booth 212 with R-SHIFT. The row is captured, becomes a ChangeEvent, the code is validated, the number becomes a pseudonym inside the HSM, the writer appends leaf 500,000 and

signs a checkpoint, tiles go to the CDN, five witnesses cosign, and Priya receives an SMS. Her phone fetches the proof and verifies it.

When a component fails

Fails	What happens
Change capture	Changes accumulate in the database log; on restart it resumes; anything older than 24 h is flagged "pending"
HSM	No new pseudonyms or signatures; a secondary HSM at another site takes over
Log writer	Changes wait in Kafka; officer terminals show "pending", not "saved"
CDN	Tiles are static and cacheable; a second CDN serves them
A witness	Three of five still suffice

Decisions

- **Who runs the log?** The Commission, with witnesses. Trust comes from who approves.
- **Tessera or Trillian?** Tessera: static tiles survive draft-day load without a live read server.
- **How many witness signatures?** Three of five.
- **Daily reconciliation or only at publication?** Both.

Build order

Sprint	Goal	Exit test
1	Log and checkpoint	A line enters, the head changes, the signature verifies
2	Intake	A deletion in the UI reaches the log with a code
3	Witness	A rewrite of an old line is caught
4	Voter	Priya gets "verified" on her phone
5	Reconciliation and appeal	A change that skipped the log is caught at night
6	Dashboard and more witnesses	No booth appears in the public feed

Storage

- **Transient:** ChangeEvent (elector number visible only up to the HSM), check requests (not stored), appeals.
- **Persisted:** tiles forever; HMAC and signing keys in the HSM; aggregate search logs for 90 days; each witness keeps every head it saw.

Closing check

One component is outside the boundary: the time source, and it has a substitute. One input looks external: whether the line is right on the ground. But it comes from the person the line is about, through the self-check. It is inside the boundary. The circle closes; and what is not promised is written explicitly: the log proves what was recorded, not that it was right.